NOT MEASUREMENT SENSITIVE

MIL-M-21861B
27 APRIL 1995
SUPERSEDING
MIL-M-21861A
3 October 1989

MILITARY SPECIFICATION

MANUAL, TECHNICAL, MATERIALS HANDLING EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the preparation and contents of technical manuals for operation, maintenance, repair and overhaul of materials handling equipment (MHE), which includes cranes, container handlers, forklift trucks, pallet trucks, tractors, platform trucks, trailers, straddle trucks, and self-propelled vehicular type equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Department of the Navy, Navy Ships Parts Control Center, Code 0341, MHE Division, 5450 Carlisle Pike, P.O. Box 2020, Mechanicsburg, PA 17055-0788, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N7123 FSC TMSS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

3.3 Binding. Manuals shall be furnished in loose leaf form in three metal ring or three metal post binders. Rings/posts shall be the type that can be opened by hand. The binder shall be rigid and durable, constructed of paperboard, fabrikoid, jute, tagboard, or synthetic material capable of withstanding normal shop usage. Manuals furnished in the loose leaf, three metal ring form, shall be provided with plastic or cardboard page protectors to prevent the first and last pages from being torn out of the manual through normal usage. Page protectors shall be a minimum of 3 inches wide by 11 inches long and shall be provided with three holes for insertion into the three ring binder. One page protector shall be placed between the front cover and the first page and another between the back cover and the last page.

3.4 Printing. The manual shall be printed in the English language by any standard commercial printing process. All printing, including tables, charts, illustrations, photographs and figures shall be clear black print and legible. Minimum quality of halftone reproduction shall be no less than a clear dot pattern. Electrostatic copies of manuals are unacceptable for quantities of ten or more. Overprinting is unacceptable. All manuals shall be a first generation print or copy of the original camera ready manual.

3.5 Cover. The information on the rigid cover shall be printed directly on the binder, printed on a full page sticker, or printed on a sheet placed within a clear plastic sleeve on the cover. Glued on stickers are unacceptable. The cover (see Figure 1) shall contain the following information:

a. A 13 digit Naval Publications and Forms Center (NPFC) stock number shall be printed at the upper right hand corner and on each odd numbered page of the manual. The NPFC stock number will be supplied by the Contracting Officer after contract award;

b. Month and year of issue of the technical manual;

c. The words "TECHNICAL MANUAL" and the statement, "OPERATION, MAINTENANCE, REPAIR AND OVERHAUL, AND ILLUSTRATED PARTS LIST";

d. Designation of equipment covered including approved MHE name, tire type (solid or pneumatic tires), type of power (electric, liquified petroleum gas, gasoline or diesel), rated lifting capacity, and the term shipboard when applicable;

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

e. Manufacturer's name and model number;

f. Testing laboratory safety rating, file number, (e.g. U.L.; type EX, EE, DS, or GS; File number XXXXXXX);

g. Contract number;

h. All of manufacturer's serial numbers;

i. All of USN registration numbers;

j. Manufacturer's logo. (Optional);

k. The following Distribution Statement shall appear at or near the bottom, "DISTRIBUTION STATEMENT C. Distribution authorized to U.S. Government agencies and their contractors. Other requests for this document shall be directed in writing to Commanding Officer, Navy Ships Parts Control Center MHE Department (Code 0341), Mechanicsburg, PA 17055-0788";

l. The contractor shall bar code each manual at extreme lower left corner of the cover below the distribution statement. Bar code shall be in accordance with MIL-STD-1189, and shall indicate the 13 digit NPFC stock number. Bar coding shall include the human-readable interpretation, and shall be printed on the cover or on a sticker adhered to the cover, and shall be protected from being scratched or damaged.

3.6 Pages. Pages shall be nominal 8-1/2 inches (216 mm) wide and 11 inches (28 mm) long with three punched holes. Pages shall be punched with three 9/32 inch (7 mm) holes plus or minus 1/32 inch (.8 mm) in diameter centered on a straight line parallel to and 11/32 inch (9 mm) plus or minus 1/32 inch (.8 mm) from the longer edge of the page. The remaining two holes shall be spaced 4-1/4 inches (108 mm) from the center hole. When material cannot be reduced for satisfactory presentation on single pages, "fold-over" pages may be used. Manuals shall have a blank page only at the end of a section. If necessary, this page shall be marked "Page intentionally left blank." Otherwise, printing shall be on both sides of the sheet. Pages shall be white paper unless otherwise specified herein. Text, figure, chart, and table orientation shall be portrait.

3.6.1 Page numbering. Pages shall be consecutively numbered within each section at the bottom of the page, with odd numbered pages on the right, and even number pages on the left. No two pages shall have the same number. Each page shall have the section number centered at the top. Alpha-numeric numbers

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

such as 1-a, 1a, 1.b shall not be used. Pages added to a section after final reproduction shall be numbered by decimals (e.g., if page 5.10 was replaced by five updated pages, they shall be numbered 5.10.1 through 5.10.5).

3.7 Paragraph, illustration, figure and table numbering. The decimal system shall be used, such as 1.1.1, 1.2.1, 3.4.1.3, etc., for numbering paragraphs, illustrations, figures, and tables. For example, 1.1 is the first paragraph in section 1, 1.1.1 is the first subparagraph in section 1, etc. The first number for each paragraph, illustration, figure, and table shall be the number of the section where it is located.

3.8 Copyright. Technical manual may be copyrighted, but no restriction shall be placed on the Government's right to reproduce the manual in whole or part for Government use.

3.9 Deficiencies/errors. Deficiencies/errors in the preliminary manual as determined by the Government shall be incorporated by the contractor in the final manual. Deficiencies/errors in the final manual as determined by the Government shall be incorporated in the manual before acceptance.

3.10 Text. The text may be arranged in one or two columns on the page or intermixed to suit the subject. Advertising will not be permitted. Abbreviations shall be kept to a minimum and shall be in accordance with MIL-STD-12.

3.10.1 Nomenclature consistency. Nomenclature shall be consistent throughout the manual. For example, a part once identified as a "cover" shall not be referred to elsewhere as a "plate". The nomenclature that is used shall agree with the parts list.

3.10.2 Specification and standards references. When Government specification or standard numbers are referred to, only the basic number shall be mentioned. References to specifications and standards shall be made as follows: "Federal Specification P-S-661", "Military Standard MIL-STD-105", etc.

3.10.3 Tables and charts. Tables and charts shall be used in conjunction with written information when it will help the user to clearly understand a process or procedure.

3.10.4 Weights and measures. The U.S. Standard Units for weights and measures shall be used with the metric conversion in parenthesis. When the truck is composed of metric components, dimensions of the metric components shall be solely metric.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

3.10.5 Temperature references. Temperature readings shall be given in degrees Fahrenheit. Degrees Centigrade, shall follow immediately in parenthesis. For example 194°F (90°C).

3.10.6 Multiple references. When items of the same figure are referenced within the same paragraph, the figure shall be listed once at the beginning of the paragraph. Any reference to items after the initial reference shall list the item number in parenthesis following the item description. For paragraphs referencing two or more figures, normal referencing procedures shall be used.

Example:

"DISASSEMBLY OF AIR VALVE (see figure 2.8)"

"Unscrew safety disc retainer (3) from valve body (1). Remove safety disc (2) and safety disc washer (4)."

3.10.7 Schematics. All hydraulic and electrical/electronic schematics shall be clear and unblemished. All schematics shall contain a legend of abbreviations and symbols which is consistent throughout the manual. If a schematic is too large to fit on an 8 1/2" X 11" sheet, a single foldout sheet may be used.

3.11 Photographs. Photographs must be clear and unblemished. Electrostatic copies must be first generation copies of the original continuous-tone print. The quality of photographs in the preliminary manual shall be an example of the quality of photographs in the final manual. If a photograph cannot be reproduced clearly, a line art picture shall be used in its place.

3.12 Format. Manual content shall appear as follows in the order shown:

.....Title Page. Refer to (3.13)

Roman numerals.....Specification Sheets. (3.14)

Roman numerals.....Environment Friendly Features (3.15)

Roman numerals.....Outline Drawing. (3.16)

Roman numerals.....Table of Contents. (3.17)

Roman numerals.....List of Illustrations. (3.18)

Roman numerals.....List of Tables and Charts. (3.19)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Section 1.....General Information. (3.20)

Section 2.....Operating Instructions. (3.21)

Section 3.....Safety Rating. (3.22)

Section 4.....Preventive Maintenance. (3.23)

Section 5.....Repair and Overhaul. (3.24)

Section 6.....Illustrated Parts Breakdown. (3.25)

Section 7.....Alpha-numeric Parts List Index. (3.26)

3.13 Title Page. The title page shall be identical to the printing on the cover of the manual (see 3.5 and Figure 1). The bar code shall not be included on the title page.

3.14 Specification Sheets. The technical manual shall include specification sheets which contain as a minimum the information referenced in figures 2 and 3 of this specification. Only applicable information shall be supplied.

3.15 Environment Friendly Features. The technical manual shall be supplied with a listing of environment friendly features which are incorporated within the forklift truck. As a minimum, the following statements shall be included, along with any other statements the manufacturer deems necessary to maintain the environmental integrity of the truck:

Painted with lead/chromate free paint by "manufacturer name", at "manufacturer facility" on "month and year".

Non-asbestos brake linings were installed by "manufacturer name", at "manufacture facility" on "date".

3.16 Outline Drawing. An outline drawing shall be supplied for the type of MHE being furnished, and shall conform to one of the outline drawings listed in figures 4 through 11 of this specification. All outline drawings shall show dimensions and weights in English units with metric units in parenthesis, unless the vehicles are entirely metric.

3.17 Table of Contents. A Table of Contents in numerical sequence shall list the major sections, paragraphs and their titles, and page numbers.

3.18 List of Illustrations. A List of Illustrations including the Illustrated Parts Breakdown shall list in numerical sequence

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

the figure number with the title and page numbers of each illustration.

3.18.1 Illustrations. Illustrations throughout the manual shall be identified by a figure number and description. An illustration requiring more than one page shall identify each sheet and the total number of sheets for that figure.

Example:

Figure 2.5. Drive Axle (Sheet 1 of X).
Figure 2.5. Drive Axle (Sheet 2 of X).

3.18.2 Illustration titles. Illustration titles shall clearly and briefly describe the illustration by giving the function or process illustrated, and/or the nomenclature of the equipment shown.

Example:

Figure 2.1 Controls, Vehicle.
Figure 2.2 Steering Gear and Attaching Parts.
Figure 2.3 Dead Man and Neutral Shift.

3.18.3 Illustration references. When a reference is made to an illustration, the reference shall be made to the figure number only. The page number shall not be used. Any reference to the items on an illustration shall be indicated with the following format: "Remove clevis pin (5, fig 1.5) that attaches the adapter (3) to frame assembly (1)."

3.18.4 Parts Listings When an illustration requires a parts listing to identify illustrated parts, it shall not be necessary to turn a page to reference the parts listing.

3.19 List of Tables and Charts. A List of Tables and Charts in numerical sequence shall have the title and page number of each table.

3.20 General Information. The General Information section shall include an introduction of the equipment, a description of the components, and the steps for preparation for use. This section shall include definitions of technical terms or uncommon terms used within the text of the manual.

3.20.1 Introduction. The introduction shall consist of a general description of the equipment, its function, and the scope of its operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

3.20.2 Description. The description shall be complete for the components, assemblies, sub-assemblies, and accessories. The illustrations and text used within this section shall give a general understanding of the physical characteristics and the function of the equipment as a whole, and shall explain the function of each control or instrument.

3.20.3 Preparation for use. The instructions describing preparation for use shall cover unpacking, assembly, lubrication, battery charging, and any other preparations, adjustments, and/or testing of the equipment before use.

3.21 Operating Instructions. The Operating Instructions shall cover all the procedures necessary for the operator to efficiently and safely use the equipment to accomplish its designated task. These operating instructions shall be sufficiently detailed to allow licensed operators not trained in this equipment to independently and safely operate the equipment without additional training or explanation. They shall describe the use of controls and instruments.

3.21.1. Notes, cautions and warnings. Notes, cautions, and warnings, shall be printed where necessary to insure proper safety and to provide warning of possible operating hazards. Warnings or Cautions should be placed in the text immediately prior to the step or procedure to which they apply.

a. NOTE: An operating procedure, condition, maintenance procedure, etc., which is essential to highlight.

b. CAUTION: Operating procedures, practices, maintenance procedure, etc., which, if not strictly observed, will result in damage to or destruction of equipment.

c. WARNING: Operating procedures, practices, maintenance procedure, etc., which will result in personal injury or loss of life if not correctly followed.

3.22 Safety Rating. Safety Rating shall be specified, such as type EX, EE, DS, GS, LPS, GS/LPS, etc. This section shall define the equipment's safety rating, include guidelines to maintaining the safety rating during maintenance, and provide instructions for an overload test.

3.22.1 Maintenance of Safety Rating. The Maintenance of Safety Rating section shall list the safeguards, special equipment, parts, and the manufacturers part numbers required to retain that rating during maintenance and repair. The contractor shall define the areas of special attention, such as the fuel system,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

exhaust components, electrical areas, etc. All pages relating to the safety rating shall be printed on green paper for easy visibility when the manual is closed. The notation, "ALL REPAIRS MUST FOLLOW THE SAFEGUARDS WITHIN THIS SECTION TO MAINTAIN THE PROPER SAFETY RATING.", shall be made at the beginning and at the end of the section, in capital letters.

3.22.2 Overload test. The manual shall include specific instructions on testing a forklift truck for 150 percent overload on the forks. If necessary, instructions shall include locating the hydraulic relief valve, changing its setting and resetting it to its original setting.

3.23 Preventive Maintenance. The Preventive Maintenance section shall include all procedures to maintain the safety rating, inspections, and maintenance procedures performed on a regular and systematic basis. This section shall include a description of the lubricants and special tools (see 3.25.3.h) required to perform preventive maintenance, and must list the requirements for the periodic, systematic and routine inspection and cleaning of assemblies, sub-assemblies, components and parts.

3.23.1 Lubricants. Each lubricant used shall be indicated by a Federal or Military specification with a crossreference to an equal commercial lubricant. A lubrication check chart shall show the point-by-point lubrication of the equipment and the lubricants to be used. The chart shall be clear and concise, and shall contain notes necessary to insure proper lubrication. Each lubrication point shall be clearly defined by an arrow on a line diagram with a reference to the proper lubricant to be used. A mechanic using the lubrication chart and diagram shall not need to reference another part of the manual or another reference.

3.23.2 Inspection Checklists. An Inspection Checklist shall specify the frequency of inspection in terms of daily, every 2 or 3 days, weekly, monthly, quarterly, semi-annually, or annually (based on a one shift operation) as applicable, or by the hours of equipment use expressed as daily, 8, 40, 100, 200, 500, 1000 or 2000 hours.

3.24 Repair and Overhaul. The Repair and Overhaul section shall cover all information necessary to locate trouble, and to complete all equipment repairs, adjustments and testing. Repair and overhaul of components of truck models not under the contract shall not be referenced in the technical manual. This section shall include the following guides, diagrams and illustrations required for repair and overhaul:

a. Guides for troubleshooting shall describe the symptoms, probable causes and instructions for corrective action;

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

b. Wiring diagrams for all electric circuits shall indicate the function of the circuits and identify the symbols used;

c. Diagrams of the complete brake, steering,and hydraulic systems shall include all the components of the system, with a reference to Section 1 for the function of each component;

d. Any illustrations, sectional views, and exploded views necessary to facilitate repair and overhaul shall be included in this section.

3.25 Illustrated Parts Breakdown.

3.25.1 Format. In the Illustrated Parts Breakdown, an illustration and its parts listing shall be located on either the same page or on facing pages of the manual. It shall not be necessary to turn a page to find a referenced part on an illustration. The illustration shall be repeated on successive pages when a parts listing is in excess of the space permitted on one page. Each illustration shall be identified by a figure number.

3.25.2 Introduction. The introduction shall include the following:

a. A legend defining figure and index codes or symbols used within the illustrated parts breakdown;

b. The information required to permit accurate and rapid understanding of the methods used in identifying the parts in the illustrated parts breakdown;

c. A Commercial and Government Entity (CAGE) list in alpha-numeric sequence, with all the names, addresses, phone numbers, and zip codes of the contractors and original equipment manufacturers within the manual.

3.25.3 Detailed requirements. The illustrated parts breakdown (See Figure 13) shall locate and identify all equipment repair parts, accessories, auxiliary components and attachments. Illustrations shall show the location of the part within the assembly. Each part of the equipment, except common hardware items such as nuts, bolts, washers, etc., shall be identified by a part number. Equipment serial numbers and nomenclature, or special codes used instead of part numbers as part identifiers are not permitted. As a minimum, the Illustrated Parts Breakdown shall include the following items:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

a. A figure and index number combination shall be used to identify an item on the parts listing. The index number shall be the item number on the figure,(e.g., Part No.2 in Figure 6.153 shall be shown as 6.153.2).

b. A prime contractor part number, and the original equipment manufacturer (OEM) part number shall be listed for all parts, except common hardware items. Each part shown on the illustration shall be identified by a single figure and index number. Contractor and OEM part numbers shall be listed in separate columns. If a part number does not exist for an item, it shall be indicated on the parts listing. The OEM shall be defined as the actual manufacturer of each part.

c. The prime contractor shall indicate the OEM part number (P/N) whenever the part has not been manufactured by either the prime contractor or his vendor.

Example:

Part	Engine fan belt
Prime contractor	P/N required
Vendor	Engine Mfg (Vendor P/N not required)
OEM	P/N required

d. The CAGE for each OEM shall be indicated in a separate column adjacent to the OEM P/N column. When the prime contractor is the OEM, the prime contractor's CAGE shall be listed.

e. Each P/N shall be identified by an item name or description assigned by the prime contractor or OEM.

f. Abbreviations such as NPN (no part number), NSS (not serviced separately), etc., shall be inserted in the applicable part number column provided such abbreviations are explained in the introduction.

g. Hardware items which are not specially designed or treated, such as bolts, screws, washers, nuts, pipe fittings, Woodruff keys, hinges, access plates, handles, etc., do not require a P/N. However, these hardware items shall be assigned a figure and index number, and must be identified by an item name or description. Descriptions shall be complete and precise for the user to be able to easily obtain replacement parts. (e.g., "Bolt, hex head, 1"-8NC x 5-1/4" lg. or washer, 2" OD x 1-1/16" ID x .177" thk.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

h. The prime contractor shall list all special tools required to perform maintenance and repair of the MHE. Special tools are those which are not included in figure 12 of this specification. Special tools shall be identified according to manufacturers and vendors part number, CAGE, description and function.

i. When several varieties of lifting heights are furnished, each lifting mechanism and related components shall be identified and grouped together as to their specific lift height.

Example:

Mast/210"
Carriage/210"
Lift cylinder/210"

Also, all components which are common to all lifting heights shall be noted and identified with a symbol designation.

3.26 Alpha-numeric Parts List Index. A complete index listing of all part numbers from all the figures in the Illustrated Parts Breakdown (section 6) shall be arranged in alpha-numeric sequence. A second column shall provide the figure and index number for each part number (see Figure 14). This list shall include all prime contractor and OEM part numbers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

comply with all the requirements of the contract. Sampling inspection, as a part of publishing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known deficient manuals, either indicated or actual, nor does it commit the Government to acceptance of deficient manuals. The Contractor shall be responsible for correction of all deficiencies in the preliminary and final manual. Deficiencies shall be as determined by the Government.

4.1.2 Inspection Criteria. The following inspection criteria are causes for rejection of the manual:

	Defect	Requirement Paragraph
a. Preliminary manual:	-Sample of technical manual is unacceptable	3.2
b. Final manual	-Separate manual for separate truck models not supplied or separate models covered in one technical manual	3.1
	-Binding Not As Specified (NAS)	3.3
	-Quality of printing NAS	3.4
	-Cover NAS	3.5
	-Page size, numbering and punched holes NAS	3.6
	-Numbering NAS	3.7
	-Text NAS	3.10
	-Photographs NAS	3.11
	-Format NAS	3.12
	-Title Page NAS	3.13
	-Specification Sheets NAS	3.14
	-Outline Drawing NAS	3.15
	-Table of Contents NAS	3.16
	-List of Illustrations NAS	3.17
	-List of Tables NAS	3.18
	-General Information NAS	3.19
	-Operating Instructions NAS	3.20
	-Safety Rating NAS	3.21
	-Preventive Maintenance NAS	3.22
	-Repair and Overhaul NAS	3.23
	-Illustrated Parts Breakdown NAS	3.24
	-Alpha-numeric Parts List Index NAS	3.25
	-Packaging NAS	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Use implied by the system

5. PACKAGING

5.1 Packaging. For shipment, technical manuals shall be individually packaged in a sealed, waterproof, transparent plastic bag.

5.2 Packing. Individual manuals shall be placed in the truck in a safe, secure location and securely fastened to the truck with tape. Manuals for bulk shipments shall be packaged in accordance with 5.1 and shall be packed in accordance with the manufacturer's commercial practices.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. These manuals will be used by military and civilian personnel to perform routine maintenance, to repair and overhaul, and to operate material handling equipment located on board ships or ashore, in temperate climates, in the arctic, antarctic or tropics, in deserts or jungles, during wartime or peacetime situations.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.

(b) Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1).

6.3 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DOD FAR Supplement 227.405-70 exempts the requirement for a DD Form 1423.

6.4 Subject term (key word) listing.

Technical manual
Material handling equipment
Safety rating

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

6.6 Prefinal Manual When a prefinal manual is listed on the Contract Data Requirements List (DD 1423), it shall be defined as the first submission of the final manual. The final manual and the prefinal manual shall be identical if no corrections or additions are found to be necessary to the prefinal manual.

Custodian:

Navy - SA
Air Force - 99

Preparing Activity:
Navy - SA

Review Activity:

Navy - SH
Army - TM
Air Force - 84

Project No.(TMSS-0302)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

0532-LP-000-0000

MAY 1994

TECHNICAL MANUAL

OPERATION, MAINTENANCE, REPAIR, AND OVERHAUL AND ILLUSTRATED PARTS LIST

TRUCK, LIFT, FORK, PRT, DIESEL, 6,000 POUND CAPACITY AT 24 INCH LOAD CENTER, SHIPBOARD

MANUFACTURER'S NAME MODEL NUMBER

SAFETY RATING - UL TYPE DS FILE NUMBER
CONTRACT NUMBER

SERIAL NUMBERS:
001 THRU 005

NAVY REGISTRATION NUMBERS:
USN 13-00001
THRU
USN 13-00005

DISTRIBUTION STATEMENT C. Distribution authorized to U.S. Government agencies and their contractors. Other requests for this document shall be directed in writing to Department of the Navy, Navy Ships Parts Control Center, MHE Division (Code 0341), 5450 Carlisle Pike, P.O. Box 2020, Mechanicsburg, PA 17055-0788.

BAR CODE
0532-LP-000-0000

FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Specification sheet

The model number, type, critical specifications, fluid capacities, dimensions (if applicable), and the original equipment manufacturer (OEM) part number shall be listed for the following component, (include any other data specified below):

1.	Engine	- nominal horsepower (see note 1) - number of cylinders - turbo (if applicable)
2.	Carburetor/air cleaner	(see note 2)
3.	Generator/alternator	- voltage (see note 3) - amperage rating
4.	Starter	- voltage (see note 4) - rating
5.	Transmission/torque converter	
6.	Drive axle assembly(s)	
7.	Radiator (see note 5)	
8.	Service brakes	- master cylinder - wheel cylinder
9.	Parking brake	
10.	Steering	- power steering pump - power steering cylinder - steer axle assembly
11.	Battery,	- refer to Figure 3 for data to be included
12.	Tires, front,	- refer to Figure 3 for data to be included
13.	Tires, rear	- refer to Figure 3 for data to be included
14.	Hydraulic system	- control valve - pump - cylinders
15.	Forks	- length - width - thickness

Notes:

(1) For electric powered equipment substitute motor(s).
(2) For diesel powered equipment substitute fuel pump.
(3) For electric powered equipment substitute controls.
(4) For electric powered equipment substitute pump motor.
(5) For electric powered equipment substitute battery connectors.

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Specification sheet

Data required for pneumatic tire

1. Service type, e.g., truck-bus, passenger car, agriculture,etc.
2. Size, e.g., P215/70*14, LT195/75*14, etc.
3. Tread, e.g., highway regular, mud-snow, all weather, etc.
4. Air retention, e.g., tube type, tubeless,etc.
5. Material, e.g.,rayon, any acceptable, etc.
6. Strength, e.g., D load range, 8 ply rating, etc.
7. Ply arrangement, e.g., bias, radial, etc
8. Maximum tire load - pounds per square inch.
9. CAGE e.g., 73842 (Goodyear), 73808 (General).
10. Manufacturer's part number and/or product code.
11. Rim Design, e.g., contour, angles, dimensions.
12. Special features - any characteristic not covered that is necessary for identification.

Tire valve data required for pneumatic tires

1. Tire and Rim Association number, e.g., TR13, TR150CW, etc.
2. Design type, e.g., straight, preformed, etc.
3. Usage Design, e.g., snap-in, clamp-in, etc.

Battery data required for electric MHE

1. Type, e.g., E, EE, or EX.
2. Voltage.
3. No. cells.
4. No. plates.
5. Ampere hour @ 6 hour rate.
6. Size (LxWxH) (In.).
7. Weight.
8. Cover, yes or no.
9. Lead length and position.
10 Connector type, e.g., EC or SB.
11. Manufacturer name and part number.
12. Date qualified to W-B-133.

Battery data required for liquified petroleum gas, gasoline and diesel powered MHE

1. Type, e.g., maintenance free.
2. Voltage.
3. Manufacturer name and part number.
4. Reserve capacity rating at 80 degrees F.
5. Cold cranking at 0 degrees F.

FIGURE 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

CRANE

Outline Drawing

a. General dimensions, including overall length, width, height with boom in travel position and capacity.

b. Wheelbase distance.

c. Outrigger spread.

d. Underclearance.

e. Size and type of tires and wheels.

f. Turning radius.

g. Minimum intersecting aisles.

h. Location of center of gravity without rated load.

i. Weight of crane (G.V.W.).

j. Boom retracted-extended dimensions.

k. Rated lifting capacities on outriggers and on rubber. Show radius, maximum boom length, and lifting area.

l. Range diagram.

m. Axle loading- with and without rated load.

FIGURE 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

STRADDLE CARRY TRUCK

Outline Drawing

a. General dimensions, including overall length, width, height and inside width, length, and height and lifting capacity.

b. Speed range both forward and reverse.

c. Maximum lift height.

d. Maximum load width, length, and height.

e. Free lift.

f. Minimum underclearance.

g. Shoe spread and length.

h. Size and type of tires and wheels. Indicate tire pressure.

i. Wheel base dimensions.

j. Turning radius, including inside and outside clearances.

k. Location of center of gravity with and without rated load.

l. Weight of truck and wheel loading with and without rated load

m. Minimum intersecting aisles.

n. Standard of safety rating (e.g., EX,EE,DS,GS,D,G,LP, etc.)

FIGURE 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

TOW TRACTOR

Outline Drawing

a. General dimensions, including overall length, width, height and capacity (drawbar pull).

b. Size and type of tires and wheels. Indicate tire pressure if applicable.

c. Wheel base dimensions.

d. Weight of truck.

e. Battery physical characteristics including voltage, cells, rated capacity and battery dimensions.

f. Battery compartment dimensions for electric powered equipment are to be noted on drawing.

g. Standard of safety rating (eg., EX, EE, DS, GS, D, G, LP, etc).

FIGURE 6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

TRUCK, AIRCRAFT CARGO LOADING

Outline Drawing

a. General dimensions, including overall length, width, overall maximum height to top of cab with platform lowered and raised and lift capacity.

b. Collapsed platform height.

c. Maximum raised platform height.

d. Platform width and length.

e. Minimum underclearance beneath truck body.

f. Size and type of tires and wheels. Indicate tire inflation pressure.

g. Wheel base dimensions.

h. Turning radius.

i. Location of center of gravity with and without rated load.

j. Weight of truck.

FIGURE 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

PLATFORM TRUCK

Outline Drawing

a. General dimensions, including overall length, width, height, height of platform and capacity.

b. Platform Width.

c. Platform Length.

d. Minimum underclearance of truck body.

e. Size and type of tires and wheels. Indicate tire inflation pressure if applicable.

f. Wheel base dimensions.

g. Turning radius.

h. Location of center of gravity with and without rated load.

i. Weight of truck (for electric powered equipment, weights with and without battery).

j. Battery physical characteristics including voltage, cells, rated capacity and battery dimension.

k. Minimum equal intersecting aisles.

l. Battery compartment dimensions for electric powered equipment are to be noted on drawing.

FIGURE 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

NATO Coordinating Committee (ASCC)

CONTAINER TOPHANDLER TRUCK

Outline Drawing

a. General dimensions of truck with tophandler, including overall length, width, height with tophandler lowered and in transport position.

b. Collapsed mast height.

c. Maximum lift height.

d. Overhead guard height.

e. Free lift.

f. Minimum underclearance beneath mast and truck body.

g. Fork spread and fork size.

h. Size and type of tires and wheels and tire pressure.

i. Wheel base dimensions.

j. Turning radius.

k. Location of center of gravity with and without rated load.

l. Weight of truck with and without tophandler.

m. Weight on drive wheels with tophandler and forks with and without rated load.

n. Weight on steer wheels with tophandler and forks with and without rated load.

o. Floor loading (PSI) with tophandler and forks with and without rated load.

p. Standard of safety rating (e.g., EX,EE,DS,GS,D,G,LP, etc.)

q. Lifting capacity with tophandler and conventional fork operation.

r. Minimum intersecting aisles.

s. Minimum right angle stacking dimensions.

t. Forward and rearward mast tilt.

FIGURE 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

FORKLIFT TRUCK
Outline Drawing

a. General dimensions, including overall length, width, height with forks lowered and lifting capacity.
b. Lifting capacity versus load center rating table or graph to include capacity rating in pounds measured 12 inches from the vertical face of forks to a zero rating at 4 inch intervals.
c. Collapsed mast height.
d. Maximum lift height.
e. Overhead guard height.
f. Free lift.
g. Minimum underclearance beneath mast and truck body.
h. Fork spread and fork thickness, length and width.
i. Size and type of tires and wheels. Tire pressure shall be indicated where applicable.
j. Wheel base dimensions.
k. Turning radius.
l. Location of center of gravity with and without rated load.
m. Weight of truck (for electric powered equipment, weight furnished should be with and without battery).
n. Battery physical characteristics for electric powered equipment including voltage, cells, rated capacity and battery dimensions if battery is provided with trucks.
o. Minimum intersecting aisles.
p. Minimum right angle stacking dimension.
q. Battery compartment dimensions for electric powered trucks are to be noted on drawing.
r. Standard of safety rating (e.g., EX,EE,DS,GS,D,G,LP, etc.)
s. Forward and rearward mast tilt.
t. Axle loading - front and rear with and without rated load.

FIGURE 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

PALLET TRUCK

Outline Drawing

a. General dimensions, including overall length, width, height with forks lowered and capacity.

b. Collapsed height.

c. Maximum lift height.

d. Minimum underclearance.

e. Fork spread and fork length, width, and thickness.

f. Size and type of tires and wheels.

g. Wheel base dimensions.

h. Turning radius.

i. Minimum right angle stacking dimension.

j. Location of center of gravity with and without rated load.

k. Weight of truck (for electric powered equipment, weight furnished should be with and without battery).

l. Battery dimensions.

FIGURE 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Contents of mechanics tool kit

Adapter, Socket Wrench
Bar, Pry
Blade, Hacksaw
Chisel and Punch Kit
Extractor Set, Screw
File, Hand
File, Hand, 6"
File, Hand, 10"
File, Hand, Metal
File, Hand, Wood
File, Thread Restorer
Finger, Mechanical
Flashlight
Frame, Hacksaw
Gage, Feeler
Gage, Sparkplug Gap
Hammer, Ball Peen
Hammer, Claw
Hammer, Plastic Face
Knife, Putty
Pin, Drift
Plastic, Hammer Face inserts
Pliers, Channel Lock
Pliers, Diagonal Cutting
Pliers, Linesman
Pliers, Slip Joint 6"
Pliers, Snap Ring.
Rule, Steel, Mechanics
Screw Driver, Flat Tip 3"
Screw Driver, Flat Tip 6"
Screw Driver, Flat Tip 9"
Screw Driver, Flat Tip 12"
Screw Driver, Offset
Screw Driver, Phillips Tip 6"
Screw Driver, Phillips Tip 12"
Screw, Starter
Scribe, Machinists
Shears, Metal Cutting R.H.
Shears, Metal Cutting L.H.
Stud Remover Set
Tape Measure, 25 Ft.
Tool Box, Portable
Wrench, Adjustable 10"
Wrench, Pipe, 10"
Wrench, Pipe, 18"
Wrench Set, Allen, English
Wrench Set, Allen, Metric
Wrench Set, Box End, English
Wrench Set, Box End, Metric
Wrench Set, Open End, English
Wrench Set, Open End, Metric
Wrench Set, Socket, English

FIGURE 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

SECTION 6

0532-LP-000-0000

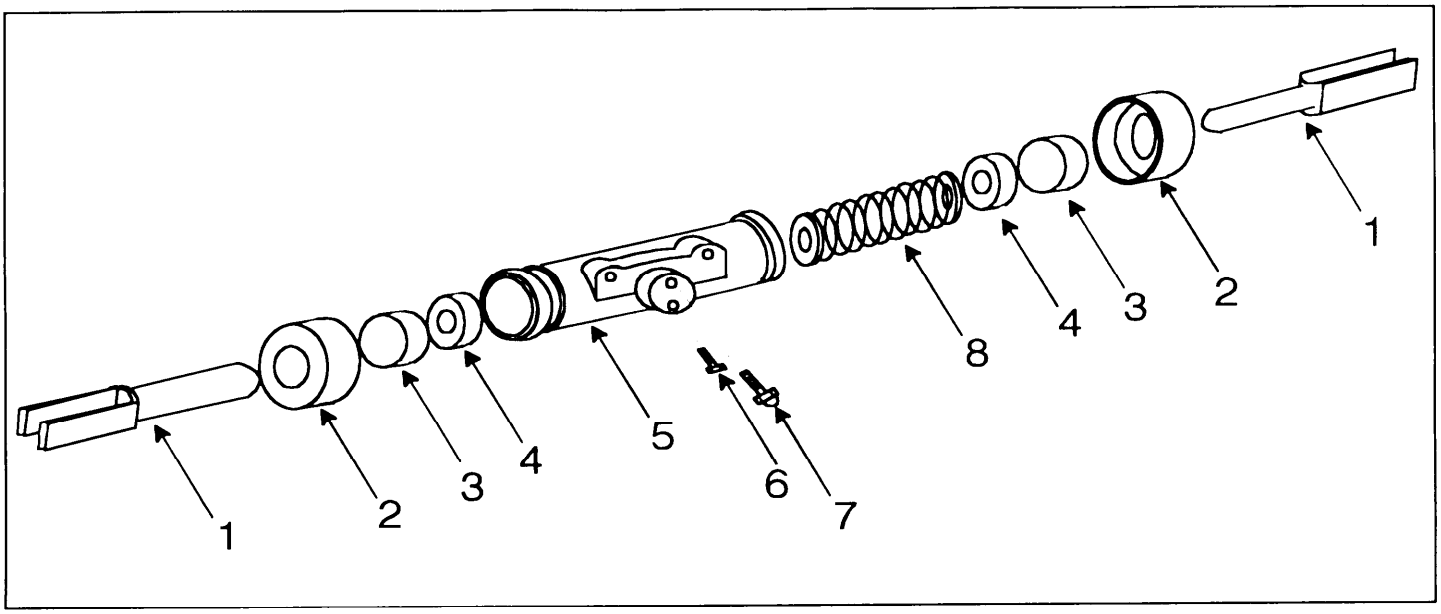


Figure 6.153. Wheel Cylinder Assembly

Figure & Index No.	Prime Contractor Part No.	OEM Part	OEM Cage	Description	QTY
6.153	935-0341	AF-690215	23455	Wheel Cylinder Assy.	
.1	927-100100	XLH-1200	11792	Actuator	2
.2	907-2112	680106	11792	Boot	2
.3	927-680106	720517	11792	Piston	2
.4	104-7904025	90125	11792	Retainer	2
.5	935-1001-31	945RB	17055	Body	1
.6	927-5150	868890	17055	Fitting, Inlet	1
.7		(NPN)		Screw, Bleeder	1
.8	935-691111	99WTD	11792	Spring	1

Example of Illustrated Parts Breakdown

FIGURE 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

PART NO.	FIGURE & INDEX NO.
AA-710-6	6.6.3
AN050075	6.1.1
A010270-15	6.1.14
A1-3761-L-90	6.153
B230089	6.18.9
B230099-1	6.123.4
B259902-FH	6.2.11
COO-55-00990	6.36.4
CZZ-33-09100	6.59.6
C1114-009	6.111.8
C222222222	6.2.22
C9-14-999	6.7.8
DA14-454545	6.124.4
DDD-ZZ-555	6.111.3
D996789	6.9.4
D996889	6.9.9
D996999	6.9.10
E1111999-09	6.30.30
E1213388-90	6.32.2
E2398098-99	6.34.5
E44-5	6.65
E44-6	6.68
E44-7	6.67
E666699999	6.53
E990902	6.123.35
FC-1539	6.153.1
FC-3208	6.153.4
FD-2230	6.153.3
FZ-987-005	6.28.6
F2-00990-34	6.99.8
F2-00990-35	6.99.3
F2-00990-38	6.99.2
F345-6789	6.66.6
F88-89-909	6.54.2
F89-89-909	6.2.17
F89-89-999	6.2.44
F909456	6.123.2
F909457	6.123.4
F909458	6.123.1
F911119	6.122.3
F997	6.166.8
F998	6.166.6
F-999-007	6.10.12
F-999-008	6.2.24
F-999-009	6.8.9
F-999-100	6.3.1
G-AA-O-1	6.76.8
GAA-PP-GSS	6.56.7
GAA-PP-LPS	6.56.12
GC-22-2134	6.45.6
MM-GOOD	6.151
M498-890	6.12.4
PP-N-555	6.77.7
ZX22-76-99	6.23.23
0114130-00	6.86.5
0133733-00	6.65.7
0981723-99	6.32.32
1099339-BB	6.9.21
123456789	6.7.8
1445628-1	6.13.13
162538090-1	6.98.2
1891023	6.56.2
2110-14	6.114
2310-14	6.115.5
2758-Z-156	6.153.2
2882-2882	6.63.21
313-212-111	6.32.1
333-364A	6.87.10
3443-A16B	6.71.3
35542A	6.122.4
3997-M-TIL	6.144.7
4004-LIT	6.44.7
41184119	6.91.1
41184120	6.91.14
41184121	6.91.20
41184122	6.91.7
41184129	6.91.2
41185001	6.91.3
41185002	6.91.8
41185004	6.91.30
41185004A	6.91.18
41185008	6.91.11
4-234-345	6.100.5
4-234-346V	6.101
555-554-553	6.21.23
89190-1	6.84.2

Example of Alpha-Numeric Parts List Index
FIGURE 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-21861 B	95/04/27

3. DOCUMENT TITLE
MANUAL, TECHNICAL, MATERIALS HANDLING EQUIPMENT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDING OFFICER NAVY SHIPS PARTS CONTROL CENTER, CODE 0341	(1) Commercial (717) 790-4025	(2) AUTOVON 430-4025
c. ADDRESS (Include Zip Code) 5450 CARLISLE PIKE MECHANICSBURG, PA 17055-0788	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete* 198-290

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.